Regulatory Reform and the Composition of FDA Drug Approvals

Evidence on Controlled Substances, Regulatory Incentives, and Pharmaceutical Market Composition

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*

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The Prescription Drug User Fee Act (PDUFA) of 1992 authorized the FDA to collect industry user fees in exchange for performance targets on review times. While a large literature documents that PDUFA substantially accelerated drug approval, no study has asked whether regulatory acceleration also changed the *composition* of approved drugs—specifically, whether the share of approvals classified as controlled substances under the Controlled Substances Act changed as a result. Using the Drugs@FDA database linked to DEA scheduling records (1939–2025, $N = 25{,}908$ original approvals), I estimate interrupted time series and difference-in-differences specifications around both PDUFA (1992) and the Generic Drug User Fee Amendments (GDUFA, 2012). The overall controlled-substance share of FDA approvals fell after PDUFA. Focusing on new drug applications (NDAs), where the policy incentive is most direct, a Poisson rate model yields an incidence-rate ratio of 1.46 ($p = 0.06$) that is not robust to group-specific trend controls. A late-2000s spike in Schedule II NDA approvals (OxyContin reformulation, Exalgo, Embeda, and related products) accounts for the apparent post-PDUFA rate increase. For GDUFA, the ANDA controlled-substance share fell significantly post-reform ($\hat{\beta} = -0.40$, IRR $= 0.67$, $p < 0.001$). The evidence does not support the hypothesis that user-fee-funded regulatory acceleration systematically increased the controlled-substance share of FDA approvals.

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1. Introduction

Drug overdose deaths reached 107,000 in the United States in 2021, the highest annual total on record, and prescription controlled substances have played a prominent role throughout the overdose epidemic (Mark and Huber 2024). Understanding how regulatory institutions shape the legal supply of controlled substances is therefore a question with first-order policy consequences.

The FDA is the primary gatekeeper of the U.S. pharmaceutical market. Its decisions determine which drugs may be sold, how quickly they reach patients, and—through the review process—what evidence of safety and efficacy is required before approval (Carpenter 2010). A large literature examines the speed-safety tradeoff at the FDA: faster review times accelerate patient access to beneficial new treatments, but deadline pressure may also reduce the scrutiny applied to drugs with later safety problems (Carpenter, Zucker, and Avorn 2008; Philipson et al. 2008). The Prescription Drug User Fee Act (PDUFA) of 1992 stands at the center of this literature: by authorizing the FDA to collect manufacturer fees in exchange for binding review-time targets, PDUFA substantially cut median approval times and changed the incentive environment facing both the agency and the firms it regulates (Ernst R. Berndt et al. 2005; M. Olson 2009).

A parallel literature studies the economics of controlled substances. Pharmaceutical opioids, stimulants, benzodiazepines, and other DEA-scheduled drugs are clinically important but carry non-trivial abuse potential. Supply-side interventions in these markets can have unintended consequences: Alpert, Powell, and Pacula (2018) show that abuse-deterrent OxyContin reformulation reduced prescription misuse but shifted users toward heroin, illustrating how the composition of the legal drug supply feeds upstream into illicit-market behavior.

What no study has examined is whether PDUFA—or its generic-drug analogue, the Generic Drug User Fee Amendments (GDUFA) of 2012—changed the composition of FDA approvals in ways that matter for controlled-substance supply. If review-time incentives disproportionately benefit high-margin drug classes, and if controlled substances systematically generate higher margins through patent-protected branded formulations, then regulatory acceleration may have altered the mix of approved drugs even without any deliberate policy intent.

This paper provides the first systematic test of that hypothesis. Using the complete Drugs@FDA database (1939–2025, $N = 25{,}908$ original approvals) linked to DEA scheduling records, I estimate interrupted time series and difference-in-differences specifications around both PDUFA (1992) and GDUFA (2012). The central finding is a null result with an important qualification. The aggregate controlled-substance share of FDA approvals fell after PDUFA, from 9.6% pre-1992 to 7.3% post-1992. A Poisson rate model restricted to new drug applications—where the PDUFA incentive is most direct—yields an incidence-rate ratio of 1.46 that is marginally significant ($p = 0.06$) under year fixed effects but collapses to 1.19 ($p = 0.62$) once group-specific time trends are added. A drug-level investigation of the 2009–2013 CS-NDA spike that drives the apparent effect reveals it to be an industry opioid product cycle (OxyContin reformulation, Exalgo, Embeda, Suboxone, and related extended-release and abuse-deterrent formulations), not a PDUFA composition effect. For GDUFA, the ANDA controlled-substance share declined significantly in the post-performance-goal era (IRR = 0.67, $p < 0.001$), which is inconsistent with the regulatory acceleration hypothesis for generic drugs.

The broader implication is that the speed-safety tradeoff literature should be extended to

downstream market-composition questions with care: the controlled-substance composition of FDA approvals is not stable over time, but the variation appears to reflect industry product cycles rather than regulatory incentives.

Roadmap. Section 2 reviews the institutional background and related literature. Section 3 describes the data and variable construction. Section 4 presents the empirical strategy. Section 5 reports the results. Section 6 interprets the findings and discusses limitations. Section 7 concludes.

2. Background and Related Literature

2.1. Institutional Background

PDUFA (1992). The Prescription Drug User Fee Act, enacted in October 1992, authorized the FDA to collect fees from pharmaceutical manufacturers to supplement congressional appropriations for drug review (Rep. Dingell 1992). In exchange, the agency agreed to binding performance goals specifying median review times for new drug applications. The original PDUFA I (1993–1997) committed the FDA to reviewing standard applications within 12 months and priority applications within 6 months of receipt, with completion targets phased in over five years. Subsequent reauthorizations—PDUFA II through VII—have progressively tightened review-time targets, expanded the user-fee program to biologics, and added post-market safety obligations. The fee structure is negotiated between the pharmaceutical industry and the FDA before each five-year reauthorization cycle, a feature that has raised concerns about regulatory capture (Darrow, Avorn, and Kesselheim 2017).

PDUFA applies to applications for new molecular entities and new formulations submitted under the NDA pathway. It does not directly govern the Abbreviated New Drug Application (ANDA) pathway for generic drugs, which was created by the Hatch–Waxman Act of 1984 (Rep. Waxman 1984). Before 2012, ANDA reviews were funded entirely by congressional appropriations, which contributed to a backlog of thousands of pending generic applications.

GDUFA (2012). The Generic Drug User Fee Amendments, signed in July 2012, extended the user-fee model to generic drug reviews under the ANDA pathway (Research Wed, 05/08/2024 - 11:53). GDUFA I (2013–2017) collected fees from generic manufacturers and established performance goals for ANDA review times. A distinguishing feature of the initial GDUFA implementation was a substantial transition period (2013–2014) during which the FDA worked to reduce a pre-existing backlog of approximately 2,500 unreviewed applications; the performance-goal era for new ANDA submissions effectively began in 2015. GDUFA II (2018–2022) and GDUFA III (2023–2027) have further refined the fee structure and review targets. Because 86.6% of controlled-substance original approvals in this dataset come through the ANDA pathway, GDUFA is arguably the more relevant policy for this paper's central question.

2.2. Theory of User-Fee Incentives

The standard economic argument for why PDUFA might alter drug composition runs through the profit motive. Under a user-fee regime with binding review-time targets, the FDA faces stronger incentives to process applications quickly. If faster review reduces the expected approval lag, it increases the net present value of approval for any drug. However, the value of accelerated approval should be highest for drugs with the largest expected revenue streams, because a shorter time-to-market compresses the revenue-loss period before patent expiration.

Controlled substances—particularly branded opioid analgesics—occupy a distinctive position in the pharmaceutical market. Many schedule II opioid formulations carry high list prices, face limited generic competition during patent protection, and generate substantial revenue for innovator firms (Powell, Pacula, and Taylor 2020). If the value of accelerated review is proportional to expected product revenue, then PDUFA may have increased the relative attractiveness of developing controlled-substance NDAs compared with other pharmaceutical product classes.

A competing hypothesis holds that the primary effect of PDUFA was to increase FDA *staffing and review resources* rather than to change the incentive environment for drug developers directly. Carpenter, Smith, et al. (2026) argue that the source of FDA funding—user fees versus congressional appropriations—matters less than the total level of review resources. Under this view, PDUFA improved review-time performance primarily by enabling the FDA to hire additional staff and retain expertise, and any composition effect would operate through the supply of review capacity rather than through demand-side incentives for manufacturers.

2.3. Related Empirical Literature

PDUFA, review speed, and approval. The most direct predecessor is Ernst R. Berndt et al. (2005), who use drug-level panel data to show that PDUFA substantially reduced NDA review times and increased approval rates, while finding little evidence of increased post-approval withdrawals. M. Olson (2009) examines launch decisions and finds that PDUFA increased the probability of U.S. initial launch conditional on development, consistent with the thesis that faster review improves the expected return to development. M. K. Olson (1997) documents substantial heterogeneity in pre-PDUFA review times across firm types, suggesting that any compositional effects may depend on which types of firms were most constrained by review delays before 1992. Dranove and Meltzer (1994) provide an early analysis of whether important drugs reach the market faster, finding limited evidence of differential queue position by therapeutic value.

Post-market safety. Carpenter, Zucker, and Avorn (2008) exploit the feature that drugs approved just before PDUFA deadlines received less review time than those approved just after, finding that proximity-to-deadline approvals are associated with higher rates of subsequent safety-related labeling changes. Darrow, Avorn, and Kesselheim (2020) provides a broad review of the literature and argues that expedited pathways have reduced clinical evidence requirements at the time of approval, raising concerns about downstream safety for certain product classes. Both

papers are informative about the speed-safety margin but do not address the market-composition question examined here.

Welfare analysis. Philipson et al. (2008) develop a formal welfare model of the FDA approval decision and use it to estimate that approval delays under the pre-PDUFA regime imposed large welfare costs. The framework emphasizes the consumer-surplus value of earlier access to beneficial drugs but does not model externalities associated with abuse potential or diversion. That omission is consequential for controlled substances, where the social cost of approval may include downstream misuse spillovers that are not internalized by approving firms or patients.

Generic drugs and GDUFA. Ernst R Berndt, Conti, and Murphy (2018) provide the first systematic economic analysis of GDUFA, documenting that the reform reduced ANDA review times and approval lags. They also analyze barriers to generic entry and argue that regulatory delays were a significant constraint on competition in the pre-GDUFA era. The analysis is primarily about generic market structure and does not focus on controlled-substance composition.

Controlled substances and opioid supply. Alpert, Powell, and Pacula (2018) show that the introduction of abuse-deterrent OxyContin reformulation in 2010 reduced prescription opioid misuse but was followed by substitution to heroin, illustrating the supply-side spillovers in controlled-substance markets. POWELL and PACULA (2021) extend this analysis to the fentanyl era. Neither paper studies the FDA approval side of the supply chain; this paper contributes by examining whether the upstream regulatory decision environment shaped the composition of what entered the market.

Gap. No existing study examines whether PDUFA or GDUFA changed the controlled-substance share of FDA approvals. The literature has studied review times, launch timing, post-market safety, and welfare under PDUFA, and market structure under GDUFA, but the composition question has not been addressed. This paper fills that gap.

3. Data

3.1. Sources

FDA Drugs@FDA. The primary data source is the Drugs@FDA flat-file database, downloaded from the FDA website in March 2026 (Research Fri, 03/13/2026 - 09:24). The database consists of twelve tab-delimited tables covering all FDA drug application and submission records from 1939 to the present. The core tables are `Applications` (one row per application, keyed by `ApplNo`), `Submissions` (one row per submission event, keyed by `ApplNo + SubmissionType + SubmissionNo`), and `Products` (product-level details linked to applications). Supporting tables cover submission classification codes, action types, document types, marketing status, and therapeutic equivalence codes.

I restrict attention to original approvals—applications with at least one submission classified as a new original application (`SubmissionClass` = ORIG or equivalent)—and collapse to one

row per application. This yields the drug-level panel of $N = 25{,}908$ observations used throughout the paper.

DEA controlled-substance reference. Controlled-substance status is determined by linking each FDA application to the DEA's official list of scheduled controlled substances (*Drug Scheduling* 2026). The linkage is ingredient-based: I match the active ingredient(s) associated with each FDA application to the DEA scheduling reference using a combination of exact-name matching, partial-string matching, and manual review. The match is conservative—only "confident scheduled" matches (ingredients with a clear one-to-one mapping to a DEA schedule) are classified as controlled substances in the primary analysis. Uncertain candidates and `List I` chemicals are excluded from the main CS indicator but appear in a broad-definition sensitivity check.

An important limitation of this approach is that the DEA linkage reflects *current* scheduling status, not historical status at the time of approval. Because scheduling can change over time, the current-status linkage may misclassify some drugs approved before they were scheduled and include others that were rescheduled after approval. The direction of the resulting bias is ambiguous but is likely small for the primary CS indicator, which focuses on commonly scheduled substances that have been on DEA schedules for decades.

3.2. Variable Construction

Outcome. The primary outcome is the annual share of original approvals that are controlled substances, constructed separately for the full sample, NDA-only, and ANDA-only subsamples. Let y_t denote the CS share in year t:

$$y_t = \frac{\text{CS approvals}_t}{\text{Total approvals}_t}$$

where "CS approvals" counts applications matched to a confident DEA schedule.

Application type. The variable `appltype_str` from the Drugs@FDA Applications table distinguishes NDA (new drug applications for brand-name and certain generic drugs), ANDA (abbreviated new drug applications for generic drugs), and BLA (biologics license applications). In the annual panel, 86.6% of CS approvals are ANDAs, reflecting the large volume of generic controlled-substance approvals—primarily opioid, benzodiazepine, and stimulant generics—processed through the ANDA pathway.

Policy timing. Three binary indicators capture the major policy transitions:

- `post_pdufa`: equals 1 for approval years 1993 and later.
- `post_gdufa`: equals 1 for approval years 2015 and later (the post-performance-goal era, excluding the 2013–2014 backlog-washout transition).
- `post_hatchwaxman`: equals 1 for approval years 1985 and later.

Event time. For event-study specifications, event time is measured relative to the policy year. For PDUFA, $\tau = t - 1992$, with the reference year set to $\tau = -1$ (1991). For GDUFA,

$\tau = t - 2012$, with reference year $\tau = -1$ (2011). End-caps are applied at $[-20, +30]$ for PDUFA and $[-10, +10]$ for GDUFA to limit sensitivity to sparse years far from the event.

3.3. Panel Structure

The drug-level panel covers 1939–2025. For the main event-study analysis, I aggregate to annual cells. The PDUFA specifications use the full 1939–2025 annual panel; the GDUFA specifications use the ANDA-only subsample covering 1984–2025. Stacked difference-in-differences specifications reshape the annual panel into a two-row-per-year format (CS group vs. non-CS group) as described in Section 4.

3.4. Summary Statistics

Table 1 reports the number of approvals and CS share by PDUFA reauthorization era. Several patterns stand out. First, the aggregate CS share is broadly stable in the 7–10% range across most eras, with a modest elevation in PDUFA II (2003–2007, 10.8%) and a sharp decline in PDUFA VII (2022–present, 4.5%). Second, ANDA volume grew dramatically after Hatch–Waxman (1984), accounting for the large increase in total approvals from the pre-Hatch–Waxman to post-Hatch–Waxman eras. Third, the NDA count has been relatively stable at 300–550 per five-year period across the PDUFA era.

Table 2 provides pre/post comparisons for PDUFA and GDUFA. The aggregate CS share fell from 9.6% pre-PDUFA to 7.3% post-PDUFA. Among NDAs, the CS share was nearly identical in the two periods (5.1% pre vs. 5.3% post). Among ANDAs, the CS share fell from 14.3% to 8.1%, driven largely by the post-Hatch-Waxman expansion in non-CS generic approvals. For GDUFA, the ANDA CS share fell from 10.8% (pre-GDUFA, $\leq$ 2012) to 7.1% (post-GDUFA performance-goal era, $\geq$ 2015).

Table 1: Original FDA Approvals by PDUFA Era

Era	Years	Total	NDA	ANDA	CS	CS Share
Pre-Hatch–Waxman	≤1984	4,270	1,488	2,005	392	9.2%
Post-HW, Pre-PDUFA	1985–1992	3,031	676	2,133	309	10.2%
PDUFA I	1993–1997	1,525	467	964	114	7.5%
PDUFA II	1998–2002	1,662	422	1,187	179	10.8%
PDUFA III	2003–2007	2,281	429	1,775	166	7.3%
PDUFA IV	2008–2012	2,761	394	2,300	210	7.6%
PDUFA V	2013–2017	3,464	551	2,830	268	7.7%
PDUFA VI	2018–2022	4,380	542	3,715	316	7.2%
PDUFA VII	2023+	2,534	308	2,113	113	4.5%
Total	1939–2025	25,908	4,277	19,022	2,067	8.0%

Source: Drugs@FDA (March 2026 extract) linked to DEA controlled-substance schedules.
CS = Controlled Substance (conservative: confident DEA scheduled matches only).
NDA counts exclude BLA (biologics) and unknown application types.

Table 2: Pre/Post Comparisons: PDUFA (1992) and GDUFA (2012)

	Total	NDA	ANDA	CS	CS/NDA	CS/ANDA
Panel A: PDUFA						
Pre-PDUFA (≤1992)	7,301	2,164	4,138	701	5.1%	14.3%
Post-PDUFA (≥1993)	18,607	3,113	14,884	1,366	5.3%	8.1%
Panel B: GDUFA (ANDA-only)						
Pre-GDUFA (≤2012)	8,359	—	8,359	858	—	10.8%
Post-GDUFA (≥2015)	7,830	—	7,830	556	—	7.1%

GDUFA transition years (2013–2014) are excluded from Panel B.
Pre-PDUFA period includes both pre-Hatch–Waxman and post-HW/pre-PDUFA years.

3.5. Limitations

Several data limitations bear on the interpretation of results. First, the DEA linkage reflects current scheduling status, not historical status at the time of approval. Second, the Drugs@FDA database is approval-centered: failed and withdrawn applications are not observed, which limits the ability to study the intensive margin of manufacturer drug development decisions. Third, the ingredient-level linkage may misclassify combination products with both controlled and non-controlled ingredients; the variable `is_multi_ingredient` flags these cases (8% of CS approvals). Fourth, review-priority coding (`review_priority`) is available but unstable over time and is not used as a main outcome. Fifth, the annual series is a single national time series without cross-sectional variation in the main specifications, limiting the credibility of causal inference.

4. Empirical Strategy

4.1. Research Design

The ideal experiment would randomly assign PDUFA or GDUFA implementation across otherwise identical regulatory environments and observe the resulting change in the controlled-substance share of approvals. In practice, both policies were enacted at single national dates and applied to all drug approvals simultaneously, so there is no cross-sectional control group in the standard sense. The analysis therefore relies on time-series comparisons—interrupted time series (ITS) and difference-in-differences (DD) designs—with pre-policy trends serving as the counterfactual for the post-policy period. I treat the estimates as policy-diagnostic first-pass evidence rather than as point-identified causal effects.

4.2. Interrupted Time Series (ITS)

The primary specification is a linear ITS estimated on the annual CS-share series:

$$y_t = \alpha + \beta_1 \cdot \text{post}_t + \beta_2 \cdot t + \beta_3 \cdot (\text{post}_t \times t) + \varepsilon_t \tag{1}$$

where y_t is the annual CS share in year t, post_t is the post-policy indicator, and the interaction $\text{post}_t \times t$ allows the post-policy trend to differ from the pre-policy trend. The coefficient β_1

estimates the level shift at the policy date; β_3 estimates the change in slope. The ITS is estimated separately for PDUFA (1993) and GDUFA (2015). For PDUFA, the sample is restricted to 1970–2025 to focus on the relevant policy window; for GDUFA, the ANDA-only series is used and 2013–2014 are excluded to isolate the backlog-washout transition.

4.3. Poisson Rate Model

Because the annual CS share is a proportion bounded between 0 and 1 and the numerator (count of CS approvals) is non-negative, the linear probability model in equation (1) may fit poorly in eras with very low or very high shares. I therefore also estimate a Poisson regression treating the annual CS count as the dependent variable and the total annual approval count as an exposure offset:

$$\log E[\mathrm{CS}_t] = \alpha + \gamma \cdot \mathrm{post}_t + \log(\mathrm{Total}_t) \tag{2}$$

where $\exp(\hat{\gamma})$ is the incidence-rate ratio (IRR) for the post-policy period relative to pre-policy, holding the total approval volume constant. Equation (2) and its variants add year fixed effects or linear time trends to absorb secular changes in the composition of approvals. All Poisson models use heteroskedasticity-robust standard errors.

4.4. Stacked Difference-in-Differences

To increase transparency about the shape of the post-policy change, I also estimate a stacked difference-in-differences specification. For each policy, I reshape the annual panel into a two-row-per-year dataset in which one observation represents CS approvals and the other represents non-CS approvals in each year:

$$\log E[n_{gt}] = \alpha_g + \lambda_t + \delta \cdot (\mathrm{CS}_g \times \mathrm{post}_t) + \log(\mathrm{Total}_t) \tag{3}$$

where $g \in \{\mathrm{CS}, \text{non-CS}\}$ indexes the group, α_g is a group fixed effect, λ_t is a year fixed effect, and δ is the DD coefficient of interest. The post indicator post_t drops out of the year-FE specification by collinearity. The Poisson DD is estimated with exposure Total_t; $\exp(\hat{\delta})$ is the IRR comparing the relative change in CS approvals to non-CS approvals after the policy. Linear (levels) and inverse hyperbolic sine (IHS) specifications are also estimated for robustness.

4.5. Event-Study DD

To examine the dynamics of any post-policy change and test for pre-policy parallel trends, I estimate an event-study version of the DD:

$$\log E[n_{g\tau}] = \alpha_g + \sum_{\tau \neq -1} \theta_\tau \cdot (\mathrm{CS}_g \times \mathbf{1}[\tau]) + \log(\mathrm{Total}_{g\tau}) \tag{4}$$

where τ is event time (calendar year relative to the policy date), and θ_τ is the group-specific coefficient at each event-time bin. End-caps are applied at $\tau \in [-20, +30]$ for PDUFA and $\tau \in [-10, +10]$ for GDUFA. The reference period is $\tau = -1$ (the year before each policy). Under

the parallel-trends assumption, $\theta_\tau = 0$ for all $\tau < 0$; a pattern of pre-policy zeros followed by post-policy deviations from zero would constitute evidence of a treatment effect (Miller 2023).

4.6. Threats to Identification

Three threats to identification are particularly relevant in this context.

Confounders contemporaneous with policy. Both PDUFA (1992) and GDUFA (2012) were enacted at times when other forces were simultaneously reshaping the pharmaceutical market. PDUFA coincided with a period of rapid expansion in ANDA volume following Hatch–Waxman, which could confound any composition effect by diluting non-CS approvals. GDUFA coincided with a surge in generic applications filed to capture expiring patents from the 2005–2012 "patent cliff." I address this by using application-type-specific subsamples (NDA-only for PDUFA, ANDA-only for GDUFA) and by including year fixed effects where applicable.

Opioid product cycle (2009–2013). A spike in Schedule II NDA approvals around 2009–2013 is visible in the data and drives the apparent post-PDUFA effect in some specifications. Section 5.3 and Section 6 document that this spike reflects a specific industry product cycle—the wave of extended-release and abuse-deterrent branded opioid formulations (OxyContin reformulation, Exalgo, Embeda, Suboxone)—rather than a sustained PDUFA-induced composition shift. The event-study design is helpful for diagnosing this: a genuine PDUFA effect should appear near $\tau = 0$ (1992–1993), not 17–19 years later.

Single time series. The national-level panel has only one observation per year, so the effective sample size for time-series inference is small (roughly 50–85 years depending on the specification). Standard errors based on heteroskedasticity alone may understate uncertainty; inference should be treated with appropriate caution.

4.7. What the Design Can and Cannot Show

This design can characterize the *reduced-form association* between PDUFA/GDUFA and the controlled-substance share of approvals. A finding of no effect would be informative: it would indicate that user-fee-funded regulatory acceleration did not systematically tilt the composition of approvals toward controlled substances. A finding of a positive effect would be consistent with the regulatory-incentive hypothesis but would not rule out confounding by contemporaneous trends.

What the design *cannot* show is whether any composition change affected downstream misuse, diversion, or overdose mortality. That would require linking the approval-side estimates to prescribing data and ultimately to health outcomes—a step outside the scope of this paper. The contribution is upstream: characterizing the approval-side composition effect (or lack thereof) is a necessary first step before studying downstream consequences.

5. Results

5.1. Descriptive Trends

Figure 1 shows the annual share of original FDA approvals classified as controlled substances from 1939 to 2025. Three features stand out. First, the share is broadly stable in the 5–15% range over the full sample, without a clear structural break at PDUFA (1992). Second, the share exhibits substantial year-to-year noise in the early part of the sample when annual approval counts were small. Third, a moderate spike is visible in the 2009–2012 period, which is investigated in Section 5.5.

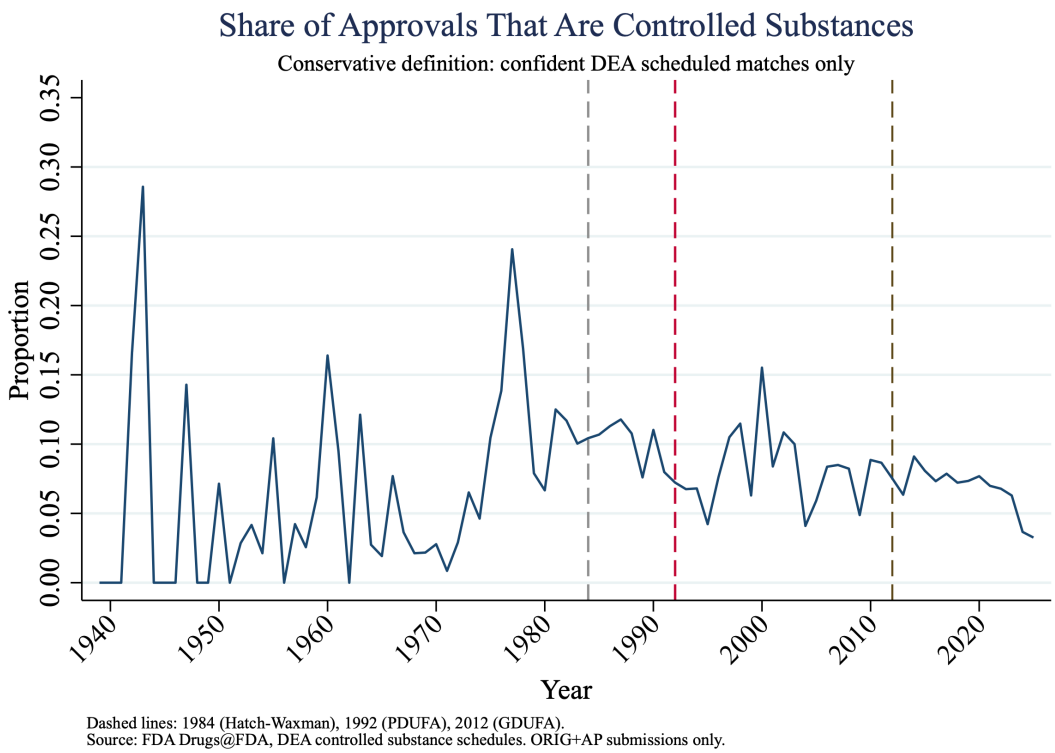

Figure 1: Annual controlled-substance share of FDA original approvals, 1939–2025. Conservative definition: confident DEA scheduled matches only. Dashed vertical lines mark 1984 (Hatch–Waxman), 1992 (PDUFA), and 2012 (GDUFA).

Figure 2 disaggregates the CS share by application type (NDA vs. ANDA). The NDA CS share fluctuates narrowly around 5–8% throughout the sample with no visible trend break at PDUFA. The ANDA CS share is higher—peaking near 20% in the 1980s—and declines after Hatch–Waxman as the expansion in non-CS generic approvals dilutes the CS share. Years with fewer than five NDA or ANDA approvals are excluded to suppress noise.

5.2. PDUFA Event Study: Full Sample

Figure 3 presents the event-study point estimates and 95% confidence intervals from the ITS specification (equation 1), with event time measured in years relative to PDUFA (1992). The pre-policy estimates ($\tau < 0$) are clustered near zero, consistent with the parallel-trends assumption.

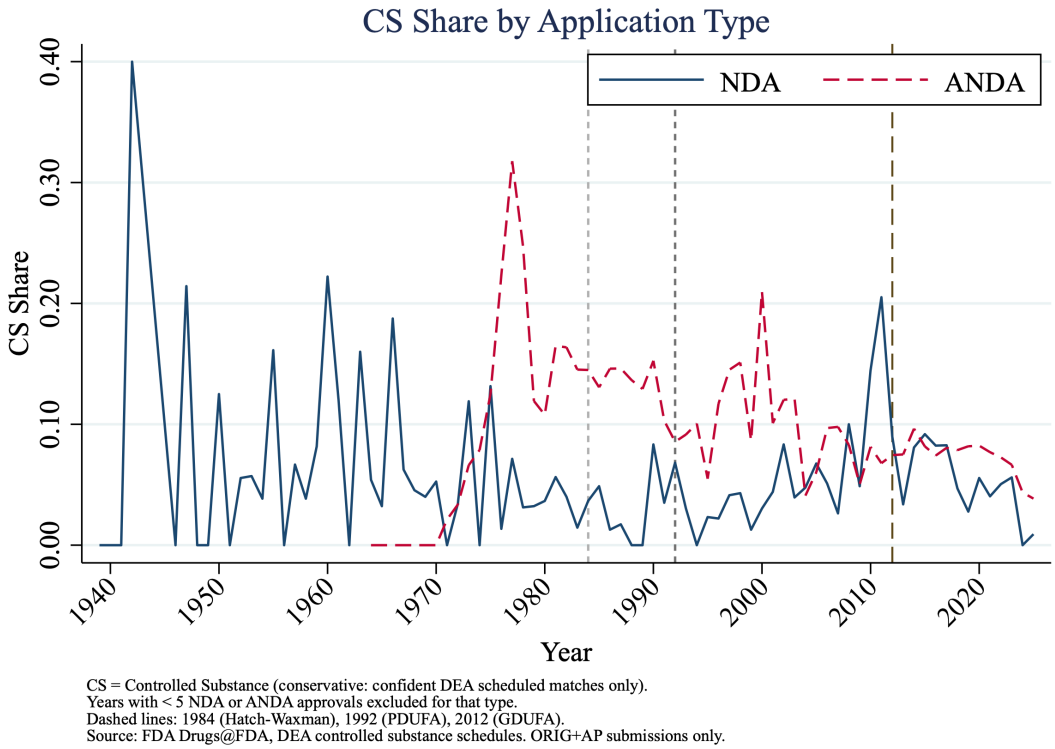

Figure 2: Annual CS share by application type (NDA vs. ANDA), 1939–2025. Years with fewer than 5 NDA or ANDA approvals are excluded. Dashed lines as in Figure 1.

Post-policy estimates ($\tau > 0$) show modest, statistically insignificant deviation from the pre-trend. There is no visible level shift or trend break at $\tau = 0$. The wide confidence intervals in both the pre- and post-periods reflect the limited power of a single national time series.

5.3. PDUFA Event Study: NDA-Only Sample

Figure 4 restricts the event-study to new drug applications (NDAs). This is the theoretically relevant subsample: PDUFA directly governed NDA review times and does not apply to ANDA submissions. The NDA CS share series is noisier because annual NDA counts are smaller, but the event-study pattern shows an important feature: the largest positive deviations from the pre-trend occur around event times 17–19 (calendar years 2009–2011), not at or near $\tau = 0$. A genuine PDUFA effect on NDA composition should be visible near the policy date.

Figure 5 shows the NDA CS share level series with pre-PDUFA and post-PDUFA mean reference lines. The post-PDUFA mean is marginally higher than the pre-PDUFA mean, but the time-series visual makes clear that the elevation is driven by the 2009–2013 period.

Table 3 reports Poisson rate models for the NDA CS count. Model (1) uses year fixed effects; Model (2) replaces year FEs with a linear time trend; Model (3) adds group-specific trends. The IRR in Model (1) is 1.46 ($p = 0.060$), suggesting a 46% higher CS approval rate post-PDUFA relative to pre-PDUFA after absorbing year-level variation. However, Model (3)—which allows pre- and post-PDUFA trends to differ by group—yields an IRR of 1.19 ($p = 0.62$), indicating that the apparent effect in Model (1) is substantially explained by differential pre- versus post-policy

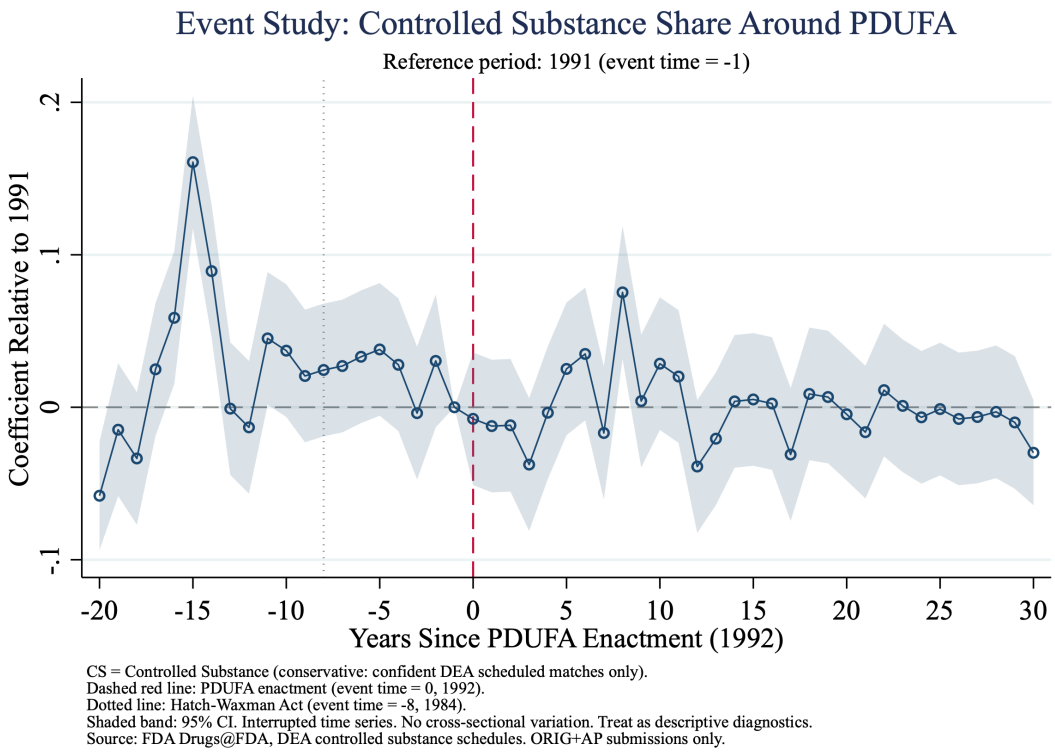

Figure 3: PDUFA event study: ITS coefficients on CS share. Event time 0 = 1992 (PDUFA enactment). Reference period: $\tau = -1$ (1991). End-caps at $[-20, +30]$. 95% confidence intervals.

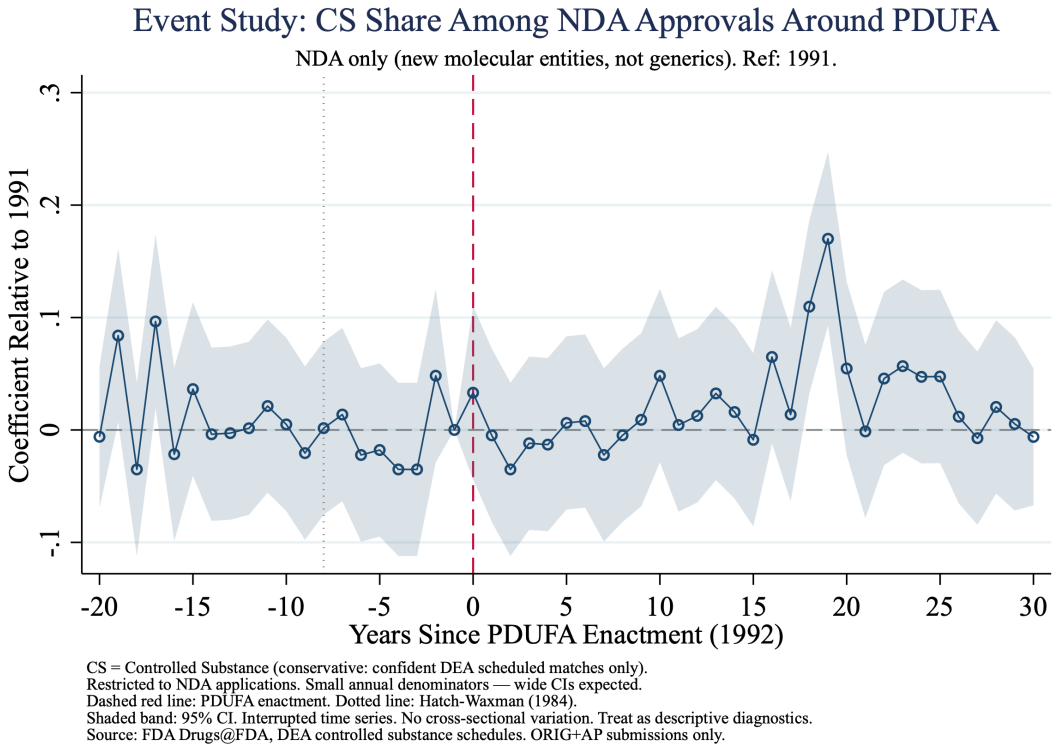

Figure 4: PDUFA event study: NDA-only sample. CS share among new drug applications. Event time 0 = 1992. Reference: $\tau = -1$.

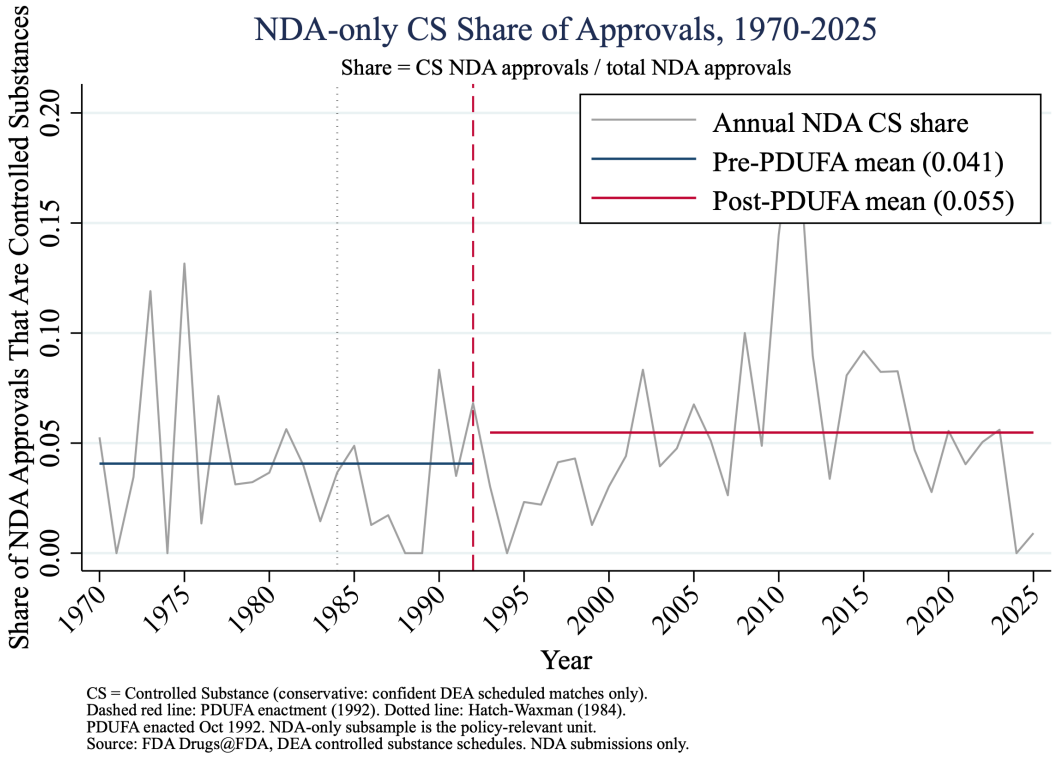

Figure 5: NDA controlled-substance share, 1970–2025, with pre/post-PDUFA mean reference lines.

trends rather than a level shift at PDUFA.

Table 3: PDUFA NDA-Only Poisson Rate Models: Sensitivity to Trend Controls

	Post-PDUFA IRR (CS vs. non-CS)		
	(1) Year FE	(2) Linear trend	(3) Group trends
IRR	1.456	1.456	1.190
z-statistic	1.879	1.832	0.490
p-value	0.060	0.067	0.624
Observations	Annual, 1970–2025, NDA only		
Exposure	Total NDA approvals		

IRR = incidence-rate ratio for post_pdufa × is_cs interaction. Robust standard errors.
Model (3) adds separate pre-policy and post-policy linear time trends by group.

Figure 6 shows the event-study DD coefficients for the NDA-only stacked Poisson specification. The IRR profile is flat and near zero for all pre-policy event times, consistent with parallel pre-trends. Post-policy IRRs drift upward gradually, reaching their largest values around event times 17–19 (2009–2011), before subsiding. This pattern—a late-rising post-policy departure that does not appear near the policy date—is inconsistent with a PDUFA effect and is instead consistent with a delayed industry product cycle, as investigated in Section 5.5.

5.4. Rate Versus Count Decomposition

Figure 7 presents a two-panel decomposition of the NDA CS result. Panel A shows the CS share of NDA approvals (the rate); Panel B shows the absolute counts of CS NDA approvals and non-CS NDA approvals over the same period. The panels illustrate that the NDA CS rate rises modestly after 1992 even as the absolute volume of non-CS NDA approvals grows faster in levels. Both patterns are simultaneously consistent: the denominator (non-CS NDA volume) growing faster than the numerator (CS NDA volume) in absolute terms does not preclude the CS rate from rising if CS NDAs grow faster in proportional terms relative to the pre-period baseline.

5.5. The 2009–2013 CS NDA Spike

The event-study and time-series evidence point to a spike in Schedule II NDA approvals concentrated in 2009–2013. Table 4 lists the CS NDA approvals from 2009 through 2013, identifying the brand name, active ingredient, sponsor, and DEA schedule for each drug. Four features stand out. First, 43 CS NDAs were approved in this five-year window, compared with an average of roughly 6–8 per year in the preceding decade. Second, 2011 alone accounts for 16 CS NDA approvals—by far the largest single-year count in the sample. Third, 25 of the 43 approvals (58%) are Schedule II, the most restrictive category for drugs with accepted medical use. Fourth, the drug list maps closely to the well-documented wave of branded extended-release and abuse-deterrent opioid formulations that characterized the late-2000s pharmaceutical market: OxyContin reformulation (Purdue Pharma, 2010), Exalgo hydromorphone (Specgx, 2010),

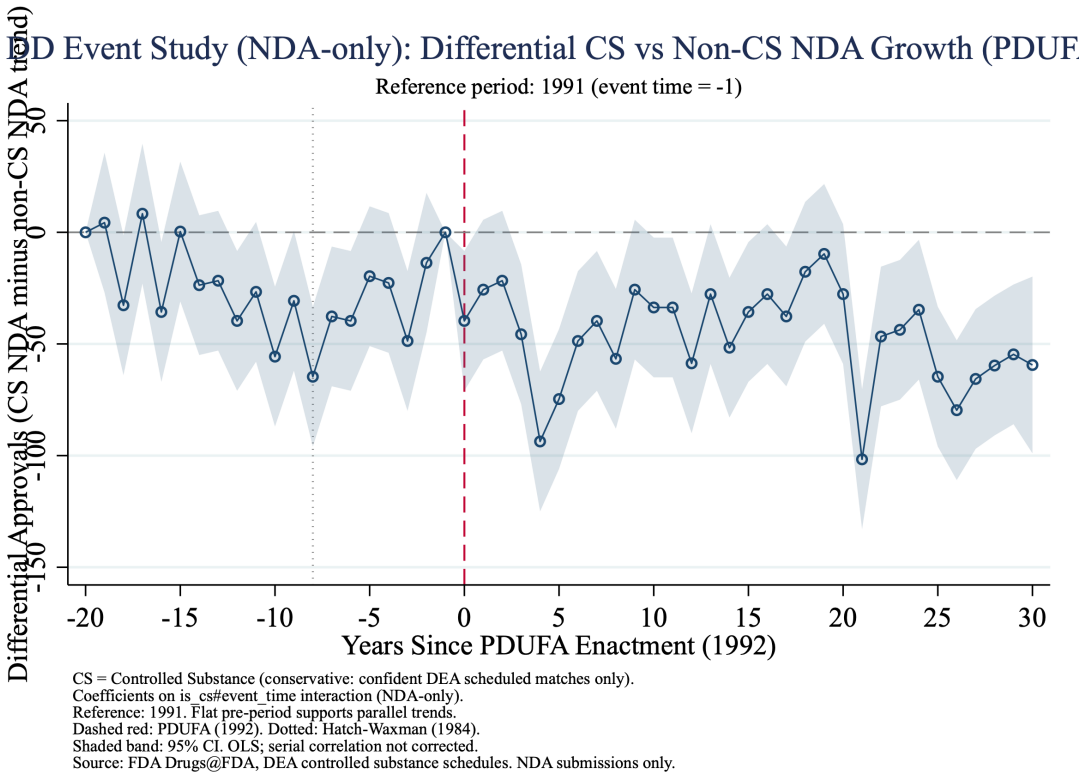

Figure 6: PDUFA NDA-only stacked Poisson DD event study. IRRs for CS $\times$ $\mathbf{1}[\tau]$ interactions. Reference: $\tau = -1$ (1991). End-caps at $[-20, +30]$. 95% CI.

Embeda morphine-naltrexone combination (Alpharma, 2009), Suboxone buprenorphine-naloxone (Indivior, 2010), and related products.

5.6. GDUFA Event Study: ANDA-Only

Figure 8 presents the GDUFA event-study DD for the ANDA-only sample. In contrast to the PDUFA result, the GDUFA event study shows a pronounced and statistically significant decline in the CS share of ANDA approvals after the post-performance-goal era begins (event times 3–13, corresponding to calendar years 2015–2025). Pre-policy event-time coefficients are clustered near zero, supporting the parallel-trends assumption over the 2002–2012 pre-GDUFA window. The stacked Poisson DD specification (Table 5) yields an IRR of 0.668 ($p < 0.001$) under year fixed effects, indicating that the CS share of ANDA approvals fell by approximately 33% relative to the non-CS share after GDUFA. This result is also robust to a linear time-trend control (IRR $= 0.668$, $p = 0.008$), though the group-specific trend specification yields an insignificant IRR of 1.38 ($p = 0.18$).

Figure 9 decomposes the GDUFA ANDA result into rate (Panel A) and count (Panel B). Panel A shows the ANDA CS share declining from a pre-GDUFA mean of 10.8% to a post-GDUFA mean of approximately 7.1%. Panel B shows that both the CS ANDA count and non-CS ANDA count declined in absolute terms after GDUFA, with the CS count falling proportionally more steeply.

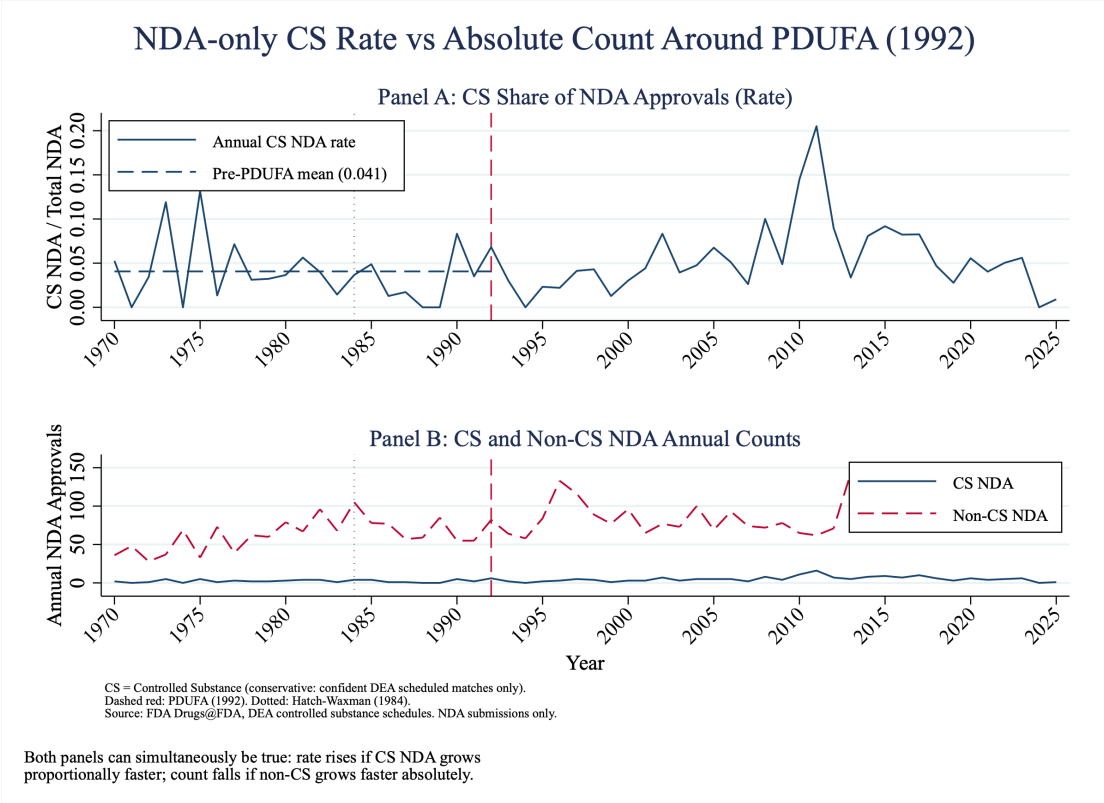

Figure 7: NDA CS approvals: rate (Panel A) and absolute count (Panel B). Panel A shows CS share among NDA approvals with pre-PDUFA mean reference line. Panel B shows annual CS NDA count vs. non-CS NDA count.

Table 4: CS NDA Approvals, 2009–2013 (Selected)

Year	Brand Name	Active Ingredient	Sponsor	Sched.
2009	ONSOLIS	Fentanyl citrate	ADALVO	II
2009	EMBEDA	Morphine sulfate; naltrexone	ALPHARMA	II
2009	CODEINE SULFATE	Codeine sulfate	HIKMA	II
2009	EDLUAR	Zolpidem tartrate	VIATRIS	IV
2010	EXALGO	Hydromorphone HCl	SPECGX LLC	II
2010	OXYCONTIN	Oxycodone HCl	PURDUE PHARMA	II
2010	SUBOXONE	Buprenorphine; naloxone	INDIVIOR	III
2010	BUTRANS	Buprenorphine	PURDUE PHARMA	III
2010	AXIRON	Testosterone	ELI LILLY	III
2010	LYRICA	Pregabalin	UPJOHN	V
2011	ANDROGEL	Testosterone	BESINS HLTHCARE	III
2011	INTERMEZZO	Zolpidem tartrate	PURDUE PHARMA	IV
2012	ZOHYDRO ER	Hydrocodone bitartrate	ZOGENIX	II
Full 2009–2013 window: 43 CS NDAs; 25 Schedule II (58%); 2011 peak: 16 approvals.				

Selected rows for illustration; full list in `output/tables/stata/cs_nda_2009_2013_drugs.csv`.
OxyContin 2010 entry is a reformulated abuse-deterrent NDA (NDA 022272), not the original approval.

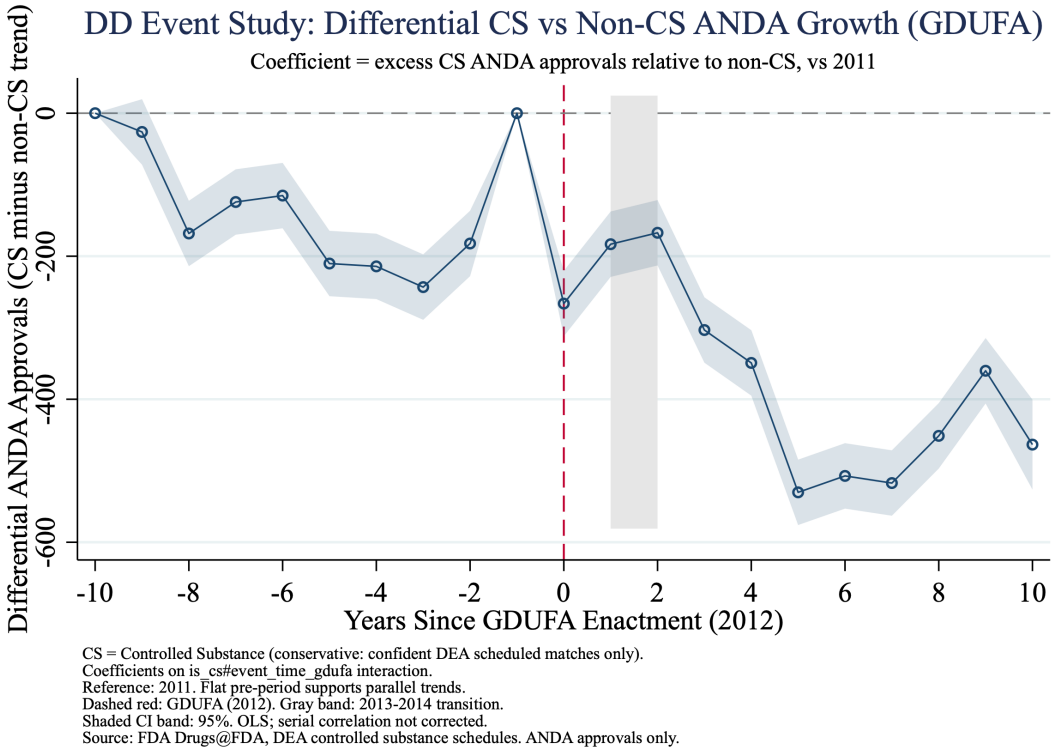

Figure 8: GDUFA ANDA-only stacked Poisson DD event study. IRRs for CS $\times$ $\mathbf{1}[\tau]$ interactions. Reference: $\tau = -1$ (2011). End-caps at $[-10, +10]$. Transition years 2013–2014 excluded. 95% CI.

Table 5: GDUFA and PDUFA Stacked Poisson DD: Main Results

	Coef.	SE	IRR	p
Panel A: GDUFA, ANDA-only (Year FE)				
post_gdufa × is_cs	−0.403	0.095	0.668	<0.001
Panel B: PDUFA, NDA-only (Year FE)				
post_pdufa × is_cs	0.376	0.200	1.456	0.060

Stacked Poisson DD with year FE and log-total-approval exposure. Robust SE.
GDUFA excludes 2013–2014 transition. PDUFA uses NDA-only sample.

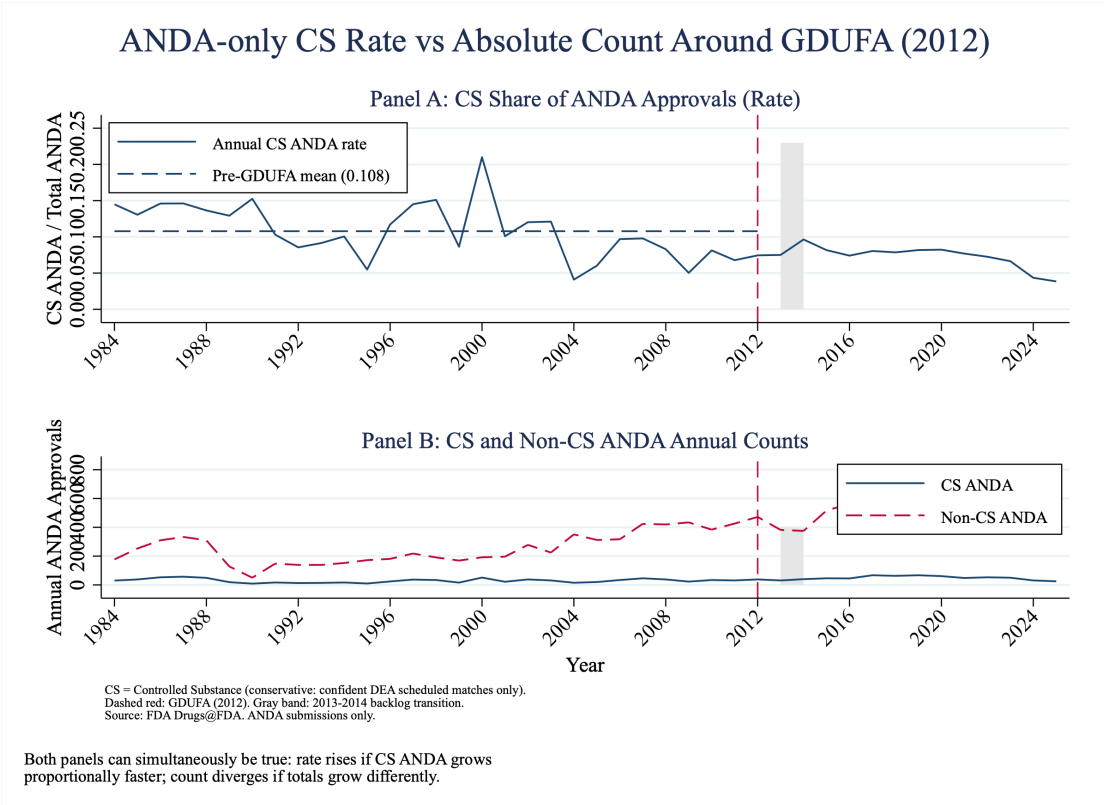

Figure 9: GDUFA ANDA CS approvals: rate (Panel A) and absolute count (Panel B). Shaded region: 2013–2014 transition period.

5.7. Sponsor Concentration

Figure 10 and Figure 11 show the Herfindahl–Hirschman Index (HHI) of sponsor concentration among CS NDA approvals (Figure 10) and all ANDA approvals (Figure 11), respectively. The NDA CS HHI is notably higher and more variable than the ANDA HHI, reflecting the smaller number of annual CS NDA approvals and the dominance of a small number of branded opioid sponsors in certain years. The GDUFA era is associated with a visible decline in ANDA CS HHI, consistent with both the entry of additional generic sponsors and the decline in CS ANDA volume.

Table 6 reports the top 10 sponsors of CS NDA approvals over the full 1970–2025 period. The market is fragmented: the leading sponsor (HIKMA and UCB INC, tied at 9 CS NDAs each) accounts for only 4.1% of the 222 CS NDAs in the 1970–2025 window. Purdue Pharma LP ranks fourth with 7 approvals (3.2%), reflecting its historical dominance in branded opioid formulations but also the relatively small absolute scale of any single sponsor's CS NDA portfolio.

6. Discussion

6.1. Interpreting the Null for PDUFA

The aggregate evidence does not support the hypothesis that PDUFA caused a sustained increase in the controlled-substance share of FDA drug approvals. The raw CS share fell after 1992. The NDA-only Poisson rate model yields a marginally significant IRR of 1.46 under year fixed

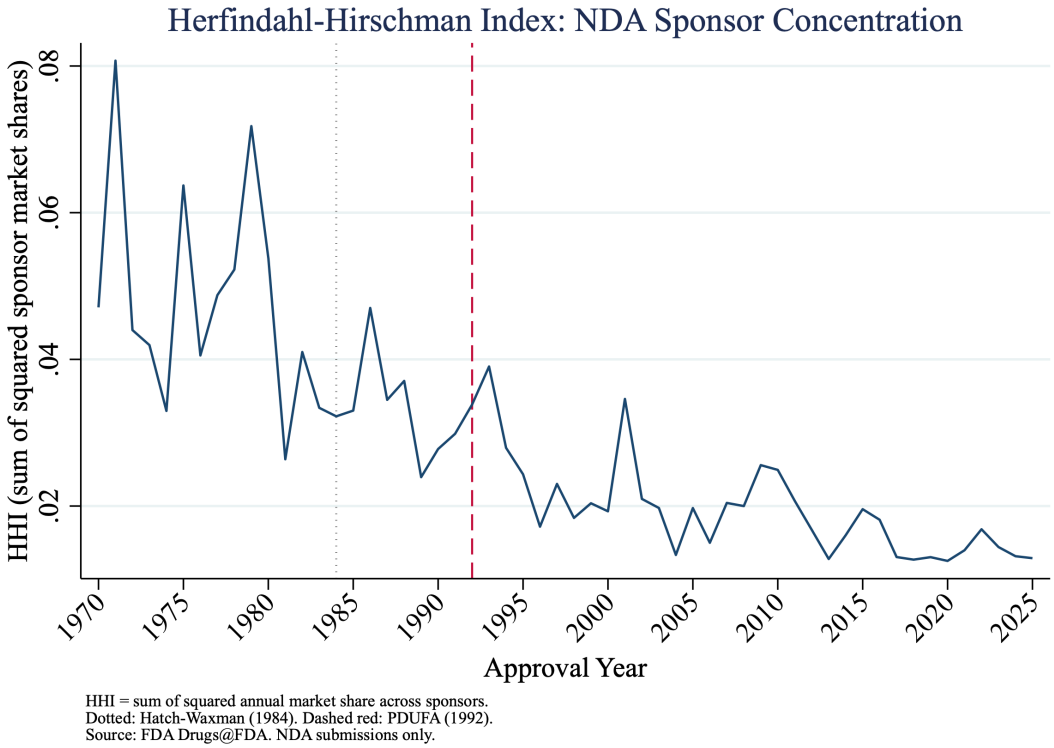

Figure 10: Sponsor HHI for CS NDA approvals, 1970–2025. Higher HHI = greater sponsor concentration. PDUFA reference lines marked.

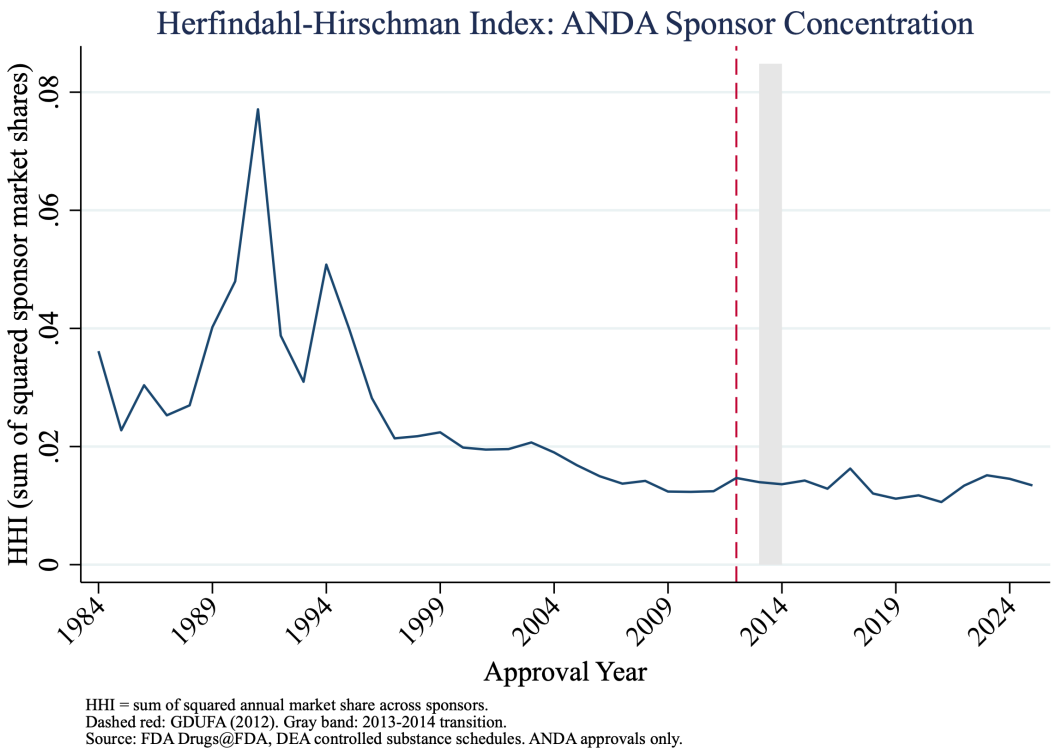

Figure 11: Sponsor HHI for ANDA approvals (all), 1984–2025. GDUFA reference line marked.

Table 6: Top 10 Sponsors of CS NDA Approvals, 1970–2025

Rank	Sponsor	CS NDAs	Share (%)
1	HIKMA	9	4.05
1	UCB INC	9	4.05
3	SPECGX LLC	7	3.15
3	PURDUE PHARMA LP	7	3.15
5	INDIVIOR	5	2.25
5	SANOFI AVENTIS US	5	2.25
5	UPJOHN	5	2.25
8	ABBOTT	4	1.80
8	COLLEGIUM PHARM INC	4	1.80
8	JANSSEN PHARMS	4	1.80
8	TAKEDA PHARMS USA	4	1.80

Universe: CS NDAs with confident DEA schedule match, 1970–2025, sponsor name non-missing.
Total CS NDAs in window: 222. Source: `cs_nda_top_sponsors_1970_2025.csv`.

effects, but this estimate is not robust: it collapses to 1.19 ($p = 0.62$) once group-specific pre- and post-policy trends are allowed. The event-study profile shows no shift at $\tau = 0$; the largest post-policy deviation occurs at $\tau \approx 17$–19 (2009–2011), far from the policy date.

These patterns are consistent with at least two non-PDUFA explanations. First, the post-Hatch-Waxman expansion in generic (ANDA) approvals after 1985 generated a large and growing pool of non-CS approvals that mechanically diluted the CS share of total approvals even if the absolute number of CS approvals continued to grow. The NDA-only restriction corrects for this composition effect at the application-type level, but even within NDAs, secular trends in the relative attractiveness of different therapeutic areas may confound inference. Second, the rise and fall of opioid pharmaceutical innovation over the 2000s and 2010s generated a specific industry product cycle that is visible in the data but is not plausibly attributable to PDUFA incentives operating 17 years later.

6.2. The 2009–2013 Opioid Product Cycle as a Separable Phenomenon

The drug-level evidence in Table 4 makes a strong case that the 2009–2013 CS NDA spike reflects the branded opioid extended-release and abuse-deterrent wave, not a PDUFA composition effect. The key features are: (1) the timing—18–21 years after PDUFA enactment—is inconsistent with an immediate regulatory incentive effect; (2) the drugs themselves are recognizable products from the documented opioid product innovation cycle, including reformulated OxyContin, Exalgo, Embeda, Suboxone, and multiple testosterone formulations; and (3) the spike subsides after 2013, suggesting a product-cycle dynamic rather than a permanent shift in approval composition.

The appropriate interpretation of this spike, for this paper's purposes, is that it is a confound: it inflates post-PDUFA IRRs in specifications without group-specific trends because the pre-PDUFA NDA CS rate was low and the spike creates a large mean difference in the post period. The fact that the group-specific-trend specification eliminates significance confirms that the spike is driving the result.

This finding is consistent with the broader opioid supply literature. Alpert, Powell, and Pacula

(2018) document that the branded opioid wave was followed by a substitution response in illicit markets; the approval-side documentation here provides complementary upstream evidence. The implication for policy is that the regulatory approval channel and the industry innovation channel are distinct: FDA approval decisions may have facilitated the opioid wave by allowing these products to come to market, but the primary driver of the wave appears to be firm-level pharmaceutical investment decisions, not the PDUFA review-time incentive structure.

6.3. Interpreting the GDUFA Decline

The GDUFA result—a 33% decline in the ANDA CS share post-2015—is the paper's most precisely estimated finding. The direction is opposite to the regulatory-acceleration hypothesis: if faster review disproportionately benefits CS generics, we would expect the CS ANDA share to rise after GDUFA. The fact that it falls is consistent with several alternative explanations.

First, the pre-GDUFA ANDA backlog may have disproportionately included CS generics filed during the 1990s–2000s opioid expansion. Clearing the backlog during 2013–2014 would have resulted in elevated CS ANDA approvals in the transition period and a subsequent reversion as the backlog drained. The three-period design (pre, transition, post-goals) partially accounts for this, but the transition-year exclusion may not fully isolate the pre-existing stock effect.

Second, the composition of ANDA applications submitted after GDUFA may have shifted toward non-CS generics, reflecting the 2012 patent cliff for non-opioid branded drugs and the subsequent filing pipeline. If non-CS generic applications surged after 2012 and began receiving approvals after the performance-goal era began in 2015, the CS share would mechanically fall even without any change in CS-specific review time or approval rates.

Third, the prescription opioid market contracted significantly after 2012 as prescribing guidelines tightened, state prescription drug monitoring programs expanded, and supply-side interventions restricted access. Reduced demand for opioid generics may have discouraged new ANDA filings for CS products relative to non-CS products.

The group-specific-trend specification yields an insignificant IRR of 1.38 for GDUFA, suggesting the same sensitivity to trend assumptions as the PDUFA result. This cautions against strong causal claims in either direction.

6.4. Limitations and Future Work

Several limitations constrain the conclusions. The most significant is the DEA linkage: because current schedules are used rather than historical schedules, any drugs that were scheduled or rescheduled after their FDA approval date are misclassified. The direction of this error for the main results is unclear. A second limitation is the single national time series, which provides limited statistical power and cannot separate PDUFA effects from contemporaneous secular trends in pharmaceutical innovation. A third limitation is the absence of a downstream outcome: this paper characterizes the approval-side composition of CS drugs but does not measure downstream prescribing volume, misuse, or diversion.

Directions for future work include: (1) constructing a historical DEA scheduling database to determine each drug's schedule as of its approval date; (2) linking approval-side CS counts to prescribing data from IQVIA or ARCOS to estimate the downstream supply implications of any

composition change; (3) extending the GDUFA analysis to GDUFA II and III as those program years accumulate; and (4) exploiting therapeutic-class or firm-level variation to identify the mechanism—review-time incentives versus aggregate staffing capacity—driving any composition effect.

7. Conclusion

This paper asks whether PDUFA (1992) and GDUFA (2012)—the two major user-fee-funded regulatory acceleration reforms at the FDA—changed the controlled-substance share of original drug approvals. Using the complete Drugs@FDA database (1939–2025, $N = 25{,}908$) linked to DEA scheduling records, I estimate interrupted time series and stacked difference-in-differences specifications for both the NDA pathway (PDUFA) and the ANDA pathway (GDUFA).

The central finding is an empirical null for PDUFA and a direction-reversing result for GDUFA. The aggregate CS share of FDA approvals fell after 1992. The NDA-only Poisson rate model yields an IRR of 1.46 that is marginally significant under year fixed effects but not robust to group-specific trend controls. A drug-level investigation confirms that the apparent post-PDUFA elevation in CS NDA approvals is driven by a specific 2009–2013 industry product cycle—extended-release and abuse-deterrent branded opioid formulations—rather than by a sustained PDUFA-induced composition shift. For GDUFA, the ANDA CS share fell significantly in the post-performance-goal era (IRR $= 0.67$, $p < 0.001$), inconsistent with the hypothesis that faster generic review disproportionately benefited CS products.

The findings suggest that the standard economic argument for why user-fee-funded regulatory acceleration should raise the CS share of approvals—that CS drugs have higher expected margins and therefore more to gain from faster review—does not find support in the approval data. The controlled-substance composition of FDA approvals appears to be driven by industry product cycles and application-mix trends rather than by review-time incentives at the regulatory margin. This does not imply that FDA regulatory policy is irrelevant to controlled-substance supply: approval decisions are a necessary condition for market entry, and any drug on the list in Table 4 required FDA clearance before it could be prescribed. But the compositional variation is better explained by demand for specific formulations in the pharmaceutical product cycle than by PDUFA incentives.

For researchers and policymakers, the implication is that the causal pathway from regulatory acceleration to controlled-substance supply likely runs primarily through the innovation investment channel—firms deciding whether to develop CS products—rather than through the review-time channel. Understanding that investment decision, and how factors like patent term, market exclusivity, and prescribing-market demand conditions shape it, may be more productive for policy design than focusing on FDA review-time performance goals. Future work linking the upstream approval-side evidence here to downstream prescribing volumes and ultimately to health outcomes would provide a more complete picture of how pharmaceutical regulation shapes controlled-substance supply across the full market chain.

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1. Research Question:

Do FDA drug-review deadlines introduced under the Prescription Drug User Fee Act (PDUFA) lead to an increased risk of post-marketing safety problems for newly approved drugs?

2. Data:

The authors compile a dataset of all FDA-approved new molecular entities (NMEs) from 1950–2004. The main analysis focuses on drugs submitted after 1993 (when PDUFA deadlines began). The data include:

- FDA approval timing
- Review cycle length
- Postmarketing safety outcomes
- Drug characteristics and therapeutic indications

Post-approval safety outcomes are measured using:

- Safety-based market withdrawals
- Addition of black-box warnings
- Discontinuation of dosage forms

Sources include Drugs@FDA, Pharmaprojects, Scrip reports, and other regulatory datasets.

3. Identification:

The key identification strategy exploits the timing of approvals relative to PDUFA review deadlines. The authors compare drugs approved in the final two months before the regulatory deadline with drugs approved:

- earlier in the review cycle
- after the deadline
- at other times in the review process

Methods include:
- Cox proportional-hazards models to study approval timing
- Exact logistic regression to estimate the probability of postmarketing safety events

The strategy tests whether drugs approved just before regulatory deadlines exhibit higher safety risks.

4. Main Results:

The introduction of PDUFA deadlines significantly changed the timing of FDA approval decisions, concentrating approvals in the two months immediately before the deadline. (p. 1359)

Drugs approved within two months before the deadline are significantly more likely to experience postmarketing safety problems:
- Safety-based withdrawal: odds ratio ≈ 5.5
- Black-box warning added later: odds ratio ≈ 4.4
- Dosage-form discontinuation: odds ratio ≈ 3.3

These safety risks remain elevated when comparing just-before-deadline approvals to both early approvals and approvals just after the deadline.

Importantly, the authors find that the increased risk is associated with the deadline itself rather than with shorter review times per se.

5. Limitations:

The study is observational, so approval timing is not randomly assigned. Unobserved factors could influence both approval timing and safety outcomes.

Measures of safety outcomes (withdrawals, warnings) capture only detected postmarket problems and may underestimate total safety risks.

The paper does not perform a full cost-benefit analysis of faster approvals versus safety risks. Other changes in drug development, regulatory practice, and monitoring over time could also affect safety outcomes.

6. How this connects to MY project:

This paper provides direct evidence that regulatory mechanisms designed to speed drug approval (PDUFA deadlines) can affect downstream safety outcomes. It demonstrates that institutional incentives within the FDA approval process can change approval timing and potentially increase the risk of postmarketing problems.

For my project on "regulatory speed and diversion risk in U.S. drug markets," this paper is important because it establishes a causal pathway from regulatory acceleration mechanisms to downstream unintended consequences. While the authors focus on safety risks (withdrawals and warnings), my research extends this logic to examine whether accelerated approval pathways may also increase diversion or misuse risks once drugs enter broader circulation.

Relevant Tags:
MUST_CITE
CORE_LIT
PDUFA
REGULATION
APPROVAL_SPEED

SAFETY_OUTCOMES
FDA_POLICY
IDENTIFICATION_IDEA, pp. 1354–1361.

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– (2020). “FDA Approval and Regulation of Pharmaceuticals, 1983-2018”. In: *JAMA*. Paper: Darrow, Jonathan J.; Avorn, Jerry; Kesselheim, Aaron S. (2020)
Title: "FDA Approval and Regulation of Pharmaceuticals, 1983–2018"
Journal: JAMA
1. Research Question:
How has the U.S. FDA drug approval system evolved from 1983 to 2018 with respect to regulatory policies, approval pathways, and the number and types of drugs approved? The paper examines how legislation, regulatory reforms, and expedited programs have shaped the speed, structure, and incentives of pharmaceutical approval in the United States.
2. Data:
The paper compiles multiple FDA administrative and regulatory datasets, including:
- FDA approvals of new drugs (1984–2018)
- Biologic approvals (1984–2018)
- Generic drug approvals (1970–2018)
- Vaccine approvals (1998–2018)
Regulatory pathway and program data:
- Orphan Drug designations (1984–2018)
- Fast Track designations (1988–2018)
- Priority Review designations (1984–2018)
- Accelerated Approval pathway (1992–2018)
- Breakthrough Therapy designations (2012–2018)
Other information includes:
- Federal legislation affecting drug regulation
- FDA funding data (especially Prescription Drug User Fee Act programs)
- Expanded access programs
- REMS requirements
The sources primarily include FDA databases, federal statutes, and regulatory reports.
3. Identification:
The paper does not use a causal identification strategy. It is a descriptive institutional analysis. The authors analyze trends in drug approvals and regulatory programs over time using descriptive statistics, policy timelines, and counts of approvals. Changes in approval patterns are discussed in relation to major legislative reforms such as the Hatch-Waxman Act (1984), the Prescription Drug User Fee Act (1992), and the FDA Safety and Innovation Act (2012), but no formal causal inference is attempted.
4. Main Results:
Several major trends emerge from the analysis.

Drug approvals fluctuated but increased again in the 2010s:
Average annual approvals were about 34 in the 1990s, fell to around 25 in the 2000s, and increased to about 41 in the 2010–2018 period.
Biologic drugs became a larger share of approvals:
Median biologic approvals rose from about 2.5 per year in the 1990s to roughly 12 per year in the 2014–2018 period.
Expedited approval programs expanded substantially:
Programs designed to accelerate drug development and review — including Fast Track, Priority Review, Accelerated Approval, and Breakthrough Therapy designation — have become increasingly common and central to the regulatory system.
FDA funding increasingly relies on industry user fees:
The Prescription Drug User Fee Act significantly increased FDA resources but also introduced review deadlines and greater interaction between regulators and pharmaceutical companies.
Generic drug approvals increased dramatically:
Following the Hatch-Waxman Act and later generic drug user fee legislation, generic approvals rose substantially, dramatically expanding the supply of generic pharmaceuticals in the U.S. market.
5. Limitations:
The study is descriptive and does not provide causal estimates of the effects of regulatory reforms on approval speed or drug outcomes. It does not analyze postmarketing safety outcomes, drug misuse, or diversion into illicit markets. The paper also does not examine firm behavior, pricing dynamics, or prescribing behavior, and therefore cannot directly assess downstream health or market consequences of regulatory changes.
6. How this connects to MY project:
This paper provides the institutional background for understanding regulatory acceleration in the U.S. pharmaceutical system. It documents the expansion of expedited approval pathways and the increasing importance of programs designed to speed drug development and regulatory review. For a project studying whether faster regulatory approval may increase diversion risk in drug markets, this paper helps establish that regulatory acceleration is a real and growing feature of the FDA system. It also shows how regulatory reforms such as PDUFA and expedited approval programs increased the speed and volume of drug approvals, which may increase legal drug supply and potentially create new pathways for misuse or diversion into illicit markets. The paper therefore provides critical context for analyzing how regulatory policy changes may generate downstream consequences beyond the intended goal of faster patient access to new drugs. (Visited on 03/08/2026).

Dranove, David and David Meltzer (1994). “Do Important Drugs Reach the Market Sooner?” In: *The RAND Journal of Economics* 25.3. Paper: Dranove, David & David Meltzer (1994) — "Do Important Drugs Reach the Market Sooner?" RAND Journal of Economics.
1. Research Question:
Do more medically or economically important drugs reach the market faster than less important drugs? Specifically, do incentives faced by pharmaceutical firms and FDA review processes lead

to shorter development and approval times for drugs with higher expected social or market value?

2. Data:

The authors construct a dataset of new drugs introduced in the United States, focusing on drugs approved by the FDA between the 1960s and 1980s. They combine:

- FDA approval data
- Patent filing dates
- Measures of drug characteristics
- Indicators of therapeutic importance

They measure "time to approval" as the time between the first worldwide patent application and FDA approval. They also construct proxies for drug importance, such as expected market size and therapeutic significance.

3. Identification:

The paper uses cross-drug variation in "importance" to estimate how it affects development and approval time. The empirical strategy is essentially a regression framework where time-to-market is the dependent variable and measures of drug importance are key explanatory variables, controlling for drug characteristics and regulatory environment. The identification relies on the assumption that variation in drug importance influences firms' development effort and regulatory prioritization but is otherwise exogenous to approval timelines.

4. Main Results:

- Drugs that are more important (economically or therapeutically) reach the market significantly faster.
- Firms appear to allocate more R&D effort to drugs with higher expected returns.
- Important drugs may also receive faster attention within the regulatory process.
- The results imply that the welfare costs of regulatory delay may be smaller than previously thought because high-value drugs tend to move through development and approval more quickly.

5. Limitations:

- Measures of "importance" rely on proxies such as market size or therapeutic category, which may imperfectly capture true social value.
- The identification strategy is observational and cannot fully separate firm behavior from regulatory prioritization.
- The analysis predates modern FDA acceleration programs (e.g., Fast Track, Breakthrough Therapy), limiting direct applicability to contemporary regulatory frameworks.
- It does not examine downstream consequences of faster approval (e.g., safety outcomes, misuse, or diversion).

6. How this connects to MY project:

This paper is foundational for understanding how incentives and regulatory processes shape the speed of drug approval. It establishes that economically or medically valuable drugs tend to move faster through development and approval pipelines. For my project on "regulatory speed and diversion risk in U.S. drug markets," this paper provides an important baseline: it shows that faster approval is not random but systematically linked to drug value and

firm incentives. My research builds on this literature by examining whether faster approval pathways may generate downstream unintended consequences — specifically, whether faster approval increases the risk of diversion or misuse in illicit markets.
Relevant Tags:
MUST_CITE
THEORY
BACKGROUND
FDA
REGULATION
APPROVAL_SPEED
PHARMACEUTICAL_ECONOMICS
PRE-PDUFA
This paper is really helpful because they create some measure and identification strategy for determining relevant indicators for measuring how important a drug is—-which is key for me and we can use a similar strategy (note that this paper is old and the regulatory landscape has changed a lot since) for developing 'importance' of a drug controls which are relevant in how fast drugs are approved or what accelerated pathway they go under" one of the earliest economic papers on FDA approval speed and it establishes a key stylized fact: **Approval speed is endogenous to drug importance.**
So in investigating whether faster FDA approval → diversion risk? it becomes hella important to consider the potential endogeniety / selection bias in any approval-speed analysis b/c this paper shows that Drug importance → faster approval, pp. 402–423. JSTOR: 2555769. (Visited on 03/08/2026).

Drug Scheduling (2026). https://www.dea.gov/drug-information/drug-scheduling. (Visited on 03/08/2026).

Mark, Tami L. and Benjamin D. Huber (Dec. 2024). "Drug Overdose Deaths Among Medicaid Beneficiaries". In: *JAMA Health Forum* 5.12, e244365. (Visited on 03/10/2026).

Miller, Douglas L. (May 2023). "An Introductory Guide to Event Study Models". In: *Journal of Economic Perspectives* 37.2, pp. 203–230. (Visited on 03/25/2026).

Olson, Mary (Jan. 2009). "PDUFA and Initial U.S. Drug Launches". In: *Michigan Telecommunications & Technology Law Review* 15.2. Paper: "PDUFA and Initial U.S. Drug Launches"
RELEVANCE TO PROJECT:
This paper studies how the Prescription Drug User Fee Act (PDUFA) affected the timing and behavior of pharmaceutical firms launching drugs in the U.S. market. Because PDUFA was designed to accelerate FDA review times, the paper provides evidence on how regulatory speed changes firm behavior and drug supply timing. This is directly relevant to a project examining whether faster regulatory approval can create downstream externalities such as drug diversion.
1. Research Question:
Did the Prescription Drug User Fee Act (PDUFA) — which increased FDA resources and accelerated review timelines — change the timing and likelihood of pharmaceutical firms launching new drugs in the United States? More broadly, the paper asks whether faster

regulatory approval alters firms' global launch strategies and incentives to bring drugs to the U.S. market sooner.

2. Data:

The paper uses data on new pharmaceutical product launches across countries, including:

- Global drug launch data
- FDA approval data
- Information on timing of drug introduction across different countries
- Drug characteristics (therapeutic class, innovation status)
- Firm characteristics

The key variable of interest is the timing of initial drug launches relative to regulatory changes introduced by PDUFA.

3. Identification:

The identification strategy exploits the introduction of PDUFA as a policy shock that accelerated FDA review times. The authors compare launch behavior before and after PDUFA implementation and analyze whether the U.S. became a more attractive early launch market relative to other countries once approval times shortened.

This effectively uses a policy change affecting regulatory speed to study how firms adjust launch timing decisions.

4. Main Results:

The paper finds that PDUFA significantly altered pharmaceutical firms' launch behavior.

Key findings include:

- FDA review times declined after PDUFA due to increased resources and review deadlines.
- Firms became more likely to launch drugs earlier in the United States relative to other markets.
- The U.S. increasingly became the first launch market for many drugs after the reform.
- Faster regulatory approval therefore shifted global launch strategies and accelerated the availability of drugs in the U.S.

5. Limitations:

The paper focuses on firm launch behavior rather than downstream outcomes. It does not study:

- prescribing patterns,
- drug misuse,
- diversion into illicit markets,
- or post-market safety outcomes.

As a result, the paper cannot assess whether faster approval generates unintended consequences such as misuse or diversion.

Additionally, the study focuses on launch timing rather than long-run drug supply or consumption patterns.

6. How this connects to MY project:

This paper provides causal evidence that regulatory acceleration policies such as PDUFA change pharmaceutical firm behavior and increase the speed at which drugs enter the U.S. market.

For a project studying whether faster regulatory approval increases diversion risk, this paper helps establish the upstream mechanism: regulatory reforms that accelerate FDA review lead firms to introduce drugs earlier and potentially increase legal pharmaceutical supply. If faster approvals expand the availability of certain drugs — especially those with abuse potential — they may create opportunities for diversion into illicit markets.
Thus, this paper contributes to the "regulatory speed → pharmaceutical supply" link that precedes the diversion mechanisms the thesis aims to study.‘, pp. 393–416.

Olson, Mary K. (1997). “Firm Characteristics and the Speed of FDA Approval”. In: *Journal of Economics & Management Strategy* 6.1.

8. Paper: Olson (1997) – Firm Characteristics and the Speed of FDA Approval

8.1. 1. Research Question:

Do firm-level characteristics influence the speed of FDA approval for new drug applications (NDAs)? Specifically, does the FDA review process vary systematically with characteristics such as firm size, R&D intensity, diversification, or experience, rather than depending only on drug characteristics?

8.1.1. 2. Data:

- Unit of observation: New Drug Applications (NDAs)
- Time period: Early 1990s (around 1990–1992)
- Sources: FDA approval data combined with firm-level data
- Key variables:
  - Outcome: FDA review time (submission → approval)
  - Drug controls: therapeutic category, novelty, importance
  - Firm characteristics: R&D intensity, firm size, diversification, experience

8.1.2. 3. Identification:

Cross-sectional regression analysis explaining variation in FDA review time.
ReviewTime_i = f(DrugCharacteristics_i, FirmCharacteristics_i)
Strategy:
Control for observable drug characteristics and test whether firm characteristics still predict approval speed. Identification relies on cross-firm variation rather than a causal policy shock (no IV or DiD).

8.1.3. 4. Main Results:

- Firms with higher R&D intensity tend to receive faster FDA approvals.
- Less diversified (more pharma-focused) firms receive faster approvals.

- Firm characteristics act as signals of credibility and expected drug quality.
- FDA review behavior appears responsive to firm reputation and expertise.

8.1.4. 5. Limitations:

- Identification is correlational; potential omitted variable bias.
- Short time window and small sample of NDAs.
- Predates major regulatory reforms such as PDUFA and modern accelerated approval programs.
- Does not study downstream effects (safety, misuse, diversion, or market outcomes).

8.1.5. 6. Connection to My Project:

This paper establishes that FDA approval speed is not random and depends partly on firm characteristics. For my project studying whether faster regulatory approval increases downstream diversion risk in U.S. drug markets, this suggests approval speed may be endogenous. Firm traits may influence both regulatory speed and eventual market supply, meaning empirical designs must account for the determinants of approval timing when estimating downstream effects.
Tags:
FDA
approval speed
regulatory behavior
firm characteristics
pharmaceutical industry
regulatory economics
BACKGROUND, pp. 377–401. (Visited on 03/08/2026).

Philipson, Tomas et al. (June 2008). "Cost-Benefit Analysis of the FDA: The Case of the Prescription Drug User Fee Acts". In: *Journal of Public Economics* 92.5. Paper:
Philipson, Tomas, Ernst Berndt, Adrian Gottschalk, and Eric Sun (2008)
"Cost-benefit analysis of the FDA: The case of the Prescription Drug User Fee Acts"
Journal of Public Economics 92: 1306–1325.
1. Research Question:
What are the welfare effects of the Prescription Drug User Fee Acts (PDUFA), which were designed to speed up FDA drug approval times?
More broadly, the paper asks how to evaluate the FDA's central policy trade-off between:
- faster access to drugs (speed of approval)
- potential safety risks (harm from unsafe drugs)
The authors develop a framework to estimate the welfare effects of regulatory speed.
2. Data:
The analysis uses several types of data:
- U.S. drug sales data
- FDA drug approval review times

- FDA drug withdrawal data (drugs later removed for safety reasons)
- Information on timing of market entry
- Data on drug patent protection and market structure

These data allow the authors to estimate the economic value of faster access to new drugs and the costs associated with potential safety risks.

3. Identification:

The empirical strategy exploits the introduction of PDUFA as a policy change that significantly reduced FDA drug review times.

PDUFA (starting in 1992) allowed the FDA to collect user fees from pharmaceutical companies in exchange for meeting review-time targets.

The authors compare outcomes before and after the policy change to estimate how faster approval times affected:

- drug availability
- producer profits
- consumer welfare
- safety risks (via withdrawal rates)

The key identification assumption is that PDUFA created a measurable exogenous increase in regulatory speed.

4. Main Results:

The authors find that PDUFA substantially reduced drug approval times and generated large welfare gains.

Key results include:

- PDUFA increased producer surplus (innovative returns) by roughly $7–$11 billion.
- Consumer welfare gains are estimated between $7–$20 billion due to faster access to new drugs.
- Faster approval allows patients to benefit earlier from new therapies.
- The welfare benefits from earlier access outweigh the expected costs associated with drug withdrawals.

Overall conclusion:

The benefits of faster access to drugs appear to exceed the costs associated with increased safety risk.

5. Limitations:

Several important limitations exist.

First, the analysis focuses on the speed-safety trade-off in terms of drug withdrawals, which is only one dimension of safety risk.

Second, the paper does not examine behavioral outcomes such as:

- prescribing patterns
- misuse
- drug diversion
- illicit market spillovers

Third, welfare calculations rely on assumptions about:

- market power during patent protection

- consumer surplus from earlier drug access
- the relationship between approval speed and safety risk.

Finally, the study evaluates aggregate welfare effects rather than distributional impacts across patient groups or drug classes.

6. How this connects to MY project:

This paper provides the foundational economic framework for thinking about the consequences of faster FDA drug approval.

Philipson et al. frame the core policy problem as a "speed vs safety" trade-off. Their analysis focuses on whether faster approval leads to harmful drugs reaching the market.

My project extends this framework by considering an additional potential externality that the paper does not examine: drug diversion and misuse.

Specifically:

Regulatory acceleration → earlier and potentially greater pharmaceutical supply → increased opportunities for diversion into illicit markets.

While Philipson et al. evaluate safety risks primarily through drug withdrawals, my project investigates whether faster regulatory approval could generate downstream effects in illicit drug markets.

Thus, this paper establishes the upstream mechanism (regulatory speed and drug supply) that motivates the empirical question in my thesis., pp. 1306–1325. (Visited on 03/08/2026).

POWELL, DAVID and ROSALIE LICCARDO PACULA (2021). "THE EVOLVING CONSEQUENCES OF OXYCONTIN REFORMULATION ON DRUG OVERDOSES". In: *American journal of health economics* 7.1, pp. 41–67. (Visited on 03/10/2026).

Powell, David, Rosalie Liccardo Pacula, and Erin A. Taylor (May 2020). "How Increasing Medical Access to Opioids Contributes to the Opioid Epidemic: Evidence from Medicare Part D". In: *Journal of health economics* 71, p. 102286. (Visited on 03/22/2026).

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– (Wed, 05/08/2024 - 11:53). "GDUFA I". In: *FDA*. (Visited on 03/27/2026).

A. Appendix

Open TODOs

No open TODOs.